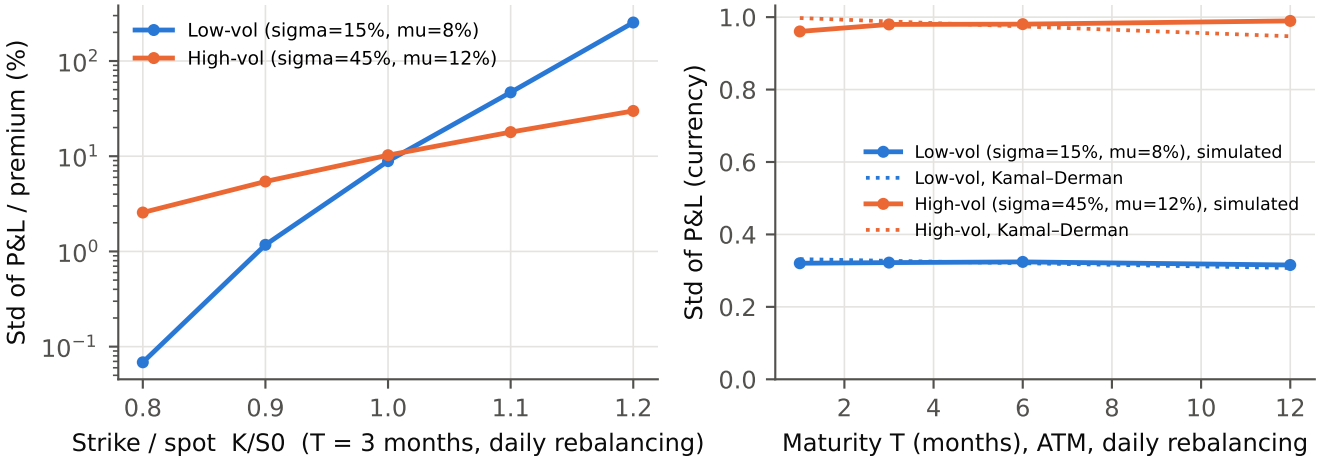
(a) Relative error explodes out of the money
Low-vol (sigma=15%, mu=8%)
High-vol (sigma=45%, mu=12%)
Std of P&L / premium (%)
10^2
10^1
10^0
10^{-1}
0.8
0.9
1.0
1.1
1.2
Strike / spot K/S0 (T = 3 months, daily rebalancing)
(b) Absolute error ~ flat in T, scales with σ
Std of P&L (currency)
1.0
0.8
0.6
0.4
0.2
0.0
Low-vol (sigma=15%, mu=8%), simulated
Low-vol, Kamal–Derman
High-vol (sigma=45%, mu=12%), simulated
High-vol, Kamal–Derman
2
4
6
8
10
12
Maturity T (months), ATM, daily rebalancing